Unsupervised Role Discovery from Professional Counter-Strike 2 Demo Data

Abstract

This project tests whether professional Counter-Strike 2 player roles can be discovered from gameplay without using role labels during training. Using 463 demos from four top-tier 2026 events, side-specific profiles were created for 214 players using combat, utility, movement, positional, and teamwork features. KMeans, Gaussian mixture models, and HDBSCAN were evaluated using internal metrics, repeated subsampling stability, and post-hoc comparison with expert role labels.

KMeans produced the clearest results. On CT-side, three clusters were recovered: Rotators, Anchors, and AWPers, with stability ARI of 0.980 and external ARI of 0.584. On T side, four clusters identified aggressive Spacetakers, a utility-oriented Spacetaker group enriched with IGLs, AWPers, and Lurkers, with stability ARI of 0.860 and external ARI of 0.461. AWPers were by far the most consistently recovered role, largely because of their distinct weapon use profile. Positional features were especially important for distinguishing CT-side rifler roles, while T-side roles depended on combinations of feature groups rather than any single category. Overall, CS2 roles are partially recoverable from aggregated demo data, but flexible rifler and IGL responsibilities remain context-dependent and overlapping.

Background

Counter-Strike 2 is a tactical team shooter in which the Terrorist (T) side attempts to plant a bomb at one of two designated sites, while the Counter-Terrorist (CT) side attempts to prevent the plant or defuse the bomb afterward. Success requires strategic planning, team coordination, mechanical skill, and rapid tactical decision-making. Over more than two decades of professional Counter-Strike, players have developed roles based on how they fight, position themselves, use utility, and support their teammates.

Although these roles are widely recognized, they are informal, overlapping, and rarely defined by a single behavior. This project investigates whether their underlying patterns can be recovered from professional match data using unsupervised machine learning. Players are clustered without access to role annotations, allowing the resulting groups to be compared afterward with established expert classifications.

Problem Statement

This project uses expert-curated annotations from NER0cs's Positions Database [4] as an external reference. These labels are used only after clustering and are treated as expert classifications rather than absolute ground truth.

The core question is:

Can professional CS2 player roles be discovered from gameplay statistics, and how closely do those clusters align with expert-curated role annotations?

Game and Role Terminology

A **demo** is a replay file containing tick-by-tick records of a match, which can reconstruct all events that occurred during the match. This includes player positions, combat events, weapon use, and grenade use that were used for this project.

In Counter-Strike, **utility** refers to grenades used to damage or blind opponents, block sightlines, and restrict movement. These include HE grenades that deal damage, flashbangs that temporarily blind players, smoke grenades that block vision, Molotovs and incendiaries that create damaging areas of fire, and decoy grenades that imitate weapon fire.

A **trade** occurs when a player eliminates an opponent who has just killed a teammate. In this project, a trade kill must occur against that opponent within five seconds of the teammate's death. Matches consist of repeated rounds, with teams switching between the CT and T sides during a match. An **opening duel** is the first elimination of a round, while **first contact** refers to the first engagement with an opponent.

The external role labels describe each player's primary responsibility rather than a specific action performed in every round. Players will have different responsibilities depending on which side they are playing on, so there are separate categories of roles for the CT and T side.

On both sides, an **AWPer** is a player that primarily uses the AWP, a long range sniper rifle typically deployed from powerful map positions to control key sightlines, secure opening picks, and create opportunities for the team. A **rifler** is a player whose primary weapons are rifles, such as the AK-47 or the M4, rather than the AWP.

An **IGL**, or in-game leader, directs team strategy and makes mid-round decisions. IGLs are also responsible for creating and working with coaches to create strategies and tactics, as well as leading the team outside of matches. Since the IGL is a leadership designation rather than a position played in game, it can overlap with any of the other categories.

On the CT-side, an **Anchor** is responsible for defending a bombsite, often remaining in a singular position to stop or delay the enemies alone when teammates rotate. On the other hand, a **Rotator** moves between areas in response to information and reinforces teammates when pressure develops elsewhere.

On the T-side, a **Spacetaker** initiates or supports contested map control and frequently participates in engagements. A **Lurker** operates separately from the main attacking group to control rotations, maintain pressure, and exploit gaps in the defense.

Pipeline

The project uses an automated data pipeline that transforms raw professional Counter-Strike 2 match demos into clustered player role profiles.

The pipeline begins with automated extraction and deduplication of professional match demos downloaded manually from HLTV.org [5]. A custom extraction script scans recent downloads, extracts valid .dem replay files, removes duplicates, and moves new demos into the project dataset directory.

Once extracted, demos are processed through a multiprocessing-based parser built on the awpy library. Awpy allows the parser to extract a wide range of gameplay information including combat events, utility usage, opening duels, trade interactions, movement behavior, and positional relationships between players.

The parsed events are transformed into side-specific player profiles using per-round combat, utility, teamwork, movement, and positional features. Selected features also include across-demo standard deviations representing behavioral consistency. All model features are standardized before clustering.

KMeans, GMM, and HDBSCAN configurations are evaluated using internal validation metrics and subsampling stability. The resulting clusters are then compared with expert role annotations and visualized using PCA projections, feature-importance charts, stability plots, and standardized feature profiles.

Interactive Dashboard

An interactive Streamlit dashboard was developed to make the results easier to explore. Users can explore individual player profiles, compare cluster assignments with expert role labels, visualize standardized features and PCA projections, and examine model performance and ablation results. The dashboard uses precomputed pipeline outputs rather than rerunning the full pipeline. The dashboard can be accessed here.

Feature Engineering

Feature engineering was a major part of the project. Since player roles in professional Counter-Strike are not directly labeled, clustering quality depended heavily on building features that captured differences in playstyle rather than just raw performance. Instead of relying only on standard statistics such as ADR (Average Damage per Round) or kills per round, the dataset includes metrics intended to describe how players take space, interact with teammates, use utility, and engage opponents.

Combat features include kills per round, survival rate, damage per round, damage taken per round, multi-kill rate, and rifle and AWP kill shares. ADR and damage differential are retained as descriptive summaries but are excluded from clustering to avoid redundant representations.

Opening duel metrics measure aggression and entry tendencies through opening kill rate, opening death rate, opening duel success rate, and first contact rate.

Trading metrics describe how players operate within their team structure. Trade kill rate, death-traded rate, and trade participation help distinguish players who remain near teammates from lurkers and other players who operate more independently.

Utility metrics record grenade usage rates, flash assists, utility damage, smoke usage, and incendiary usage per round.

Finally, positional metrics are computed directly from replay tick data and describe where players spend their time relative to teammates and opponents. These include average distance to enemies, distance to the team centroid, nearest teammate distance, movement distance per round, time spent near enemies, and stationary behavior rate.

Clustering Methodology

KMeans, Gaussian Mixture Models (GMM), and HDBSCAN were each included because they capture different types of structure within the data. KMeans partitions players into compact clusters based on centroid distance, while GMM models clusters probabilistically, allowing softer boundaries between overlapping playstyles, which may be useful due to the sometimes flexible nature of the roles. HDBSCAN was included to test whether the data contained density-based groups or outlier profiles without forcing every player into a cluster.

Before clustering, all numeric features were standardized using StandardScaler so that features with larger numeric ranges would not dominate the clustering process. A Principal Component Analysis (PCA) was also computed for visualization purposes, allowing high-dimensional player profiles to be projected into two dimensions.

A grid search for each of the models was used. KMeans and GMM were evaluated with k values of {3,4,5,6,7,8}. HDBSCAN was evaluated with minimum cluster sizes in {2,3,4,5,6,8,10,12,15} and minimum-samples values in {1,2,3,4,5,6,7,8}. HDBSCAN solutions assigning more than 30% of players to noise were excluded.

Models were ranked by a composite score combining silhouette score (40%), an inverted Davies-Bouldin index (20%), and subsampling stability (40%), where the stability term was penalized by its own standard deviation to discount models whose cluster assignments had high variability across resamples. Silhouette score measures how well-separated clusters are, while Davies-Bouldin index measures cluster compactness and overlap. However, these metrics alone may produce misleading results in high-dimensional data.

To address this, the project uses a subsampling stability analysis based on ARI, that we will call "stability ARI". For each clustering configuration, 80% of the player-side profiles are repeatedly resampled without replacement and reclustered. The resulting cluster assignments are then compared against the original clustering solution using ARI, measuring how consistently the discovered structures reappear.

Results and Analysis

CT Side

The CT-side results are best represented by the three-cluster KMeans solution. The model produced three interpretable clusters corresponding broadly to Anchors, Rotators, and AWPers. It achieved a silhouette score of 0.149, a Davies-Bouldin index of 2.195, and a composite model-selection score of 0.504. Its mean subsampling stability ARI is 0.980 ± 0.027. This high stability indicates that the CT cluster structure is reproduced consistently when the player pool changes. Of the 192 eligible CT-side player profiles, 128 were matched to the external CT-specific role labels. Against these matched labels, the three-cluster KMeans solution achieves an external ARI of 0.584 and an overall matched-sample purity of 78.1%. The three clusters have the following interpretations.

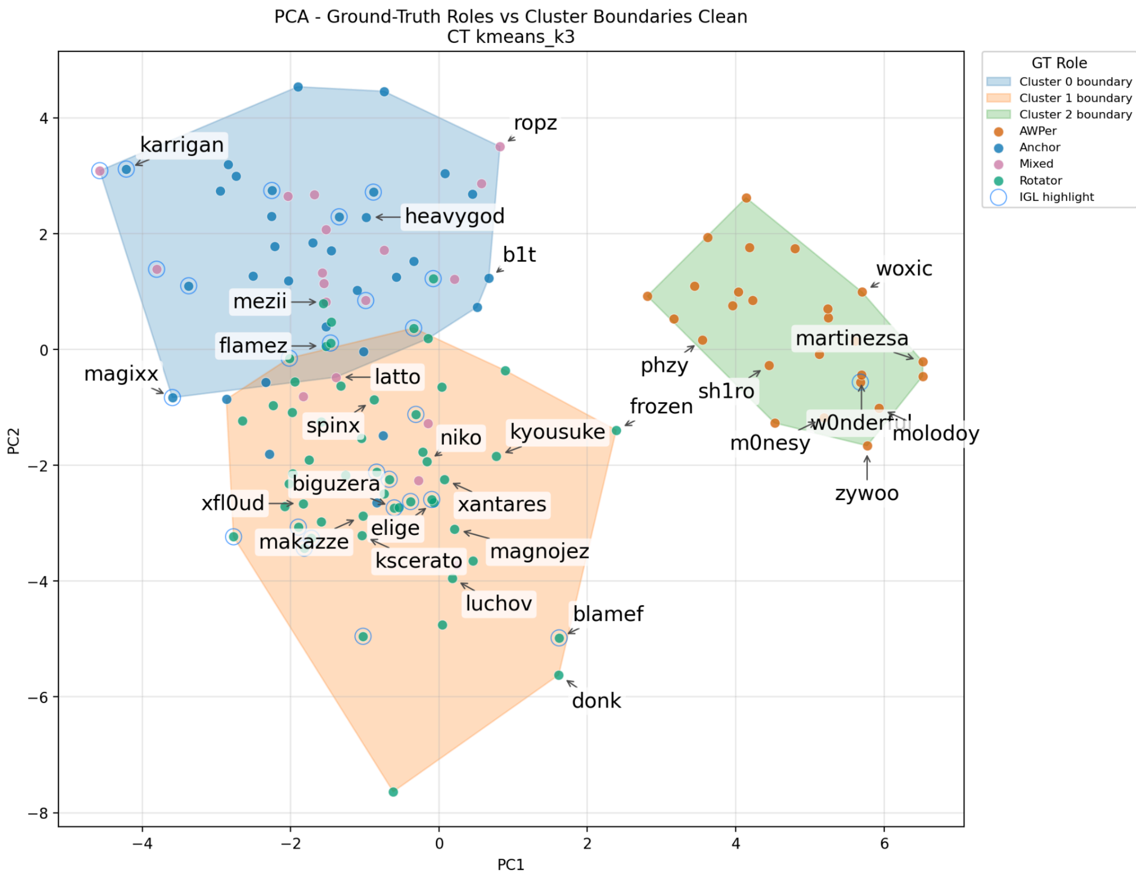


Figure 1. PCA projection of CT-side kmeans_k3 solution and external roles for matched players. Notable players are labeled.

Cluster 0 - Anchors, 58.3% purity: This cluster contains 74 players, with 48 matched to the external labels. Twenty-eight of those matched players were labeled as Anchors. The lower purity suggests that Anchor behavior is less separable than Rotator behavior, potentially because anchoring varies by map, bombsite, opponent, and team system. Some Anchors hold isolated positions and see little early action, while others are frequently pressured or play more active defensive setups. The cluster therefore represents an Anchor-like profile but includes riflers with overlapping behaviors.

Cluster 1 - Rotators, 85.5% purity: This cluster contains 80 players, of whom 55 were matched to the external labels. Forty-seven of the matched players are labeled as Rotators. It is the cleanest non-AWPer CT cluster and suggests that Rotators have a relatively consistent behavioral profile. The model-level feature importance shows that distance to the team centroid, relative team-centroid distance, nearest-teammate distance, distance to enemies, time near enemies, and first-contact behavior contribute most to the CT partition. These variables are consistent with a role that involves constant repositioning, reinforcing teammates, and responding to pressure across the map according to round state and information.

Cluster 2 - AWPers, 100.0% purity: This cluster contains 38 players, of whom 25 were matched to the external CT labels. All 25 matched players are labeled as AWPers. The cluster is defined most clearly by weapon-profile variables, particularly AWP kill share and rifle kill share, although positional and contact-related variables also contribute significantly.

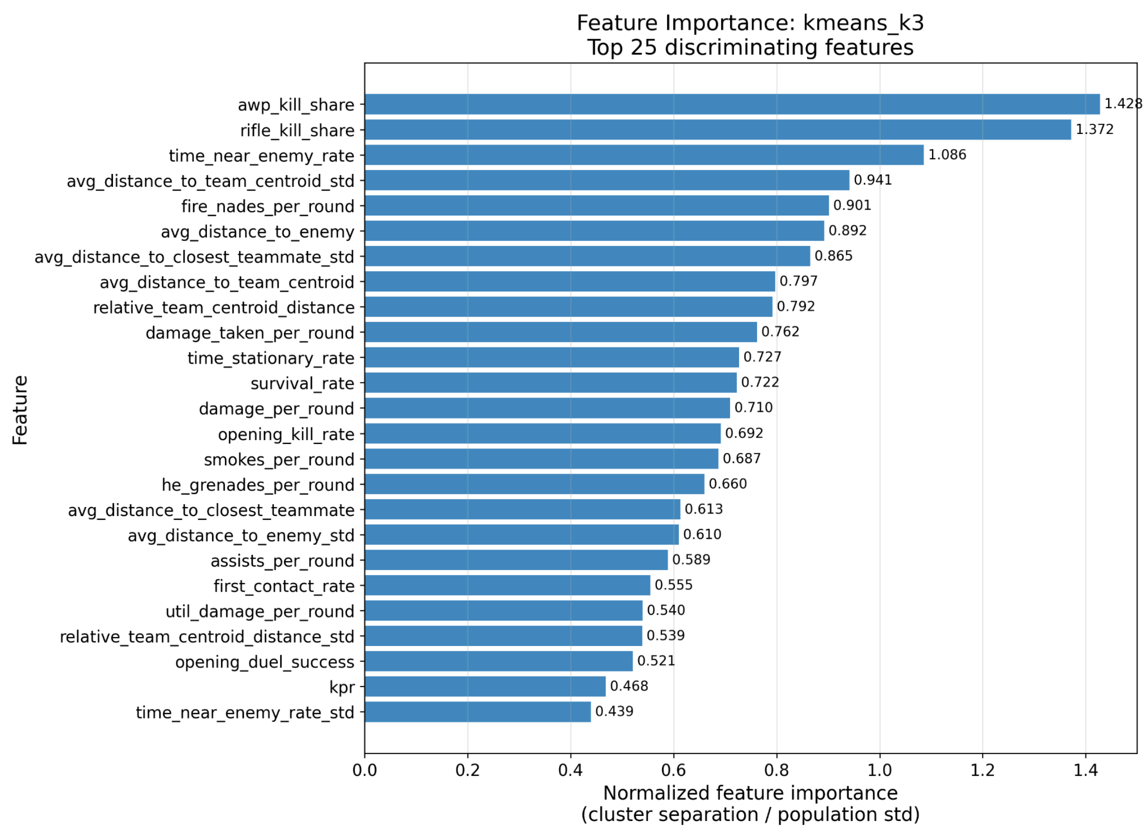


Figure 2. Top 25 features distinguishing the CT-side kmeans_k3 clusters.

The model does not recover a distinct CT-side IGL cluster. Its ARI against the binary IGL labels is -0.024, and its balanced accuracy is 0.500, indicating no meaningful separation beyond the majority-class baseline. CT-side IGLs are instead distributed among the Anchor, Rotator, and AWPer clusters. This is consistent with IGL being a leadership responsibility rather than a single mechanical role. An IGL may anchor, rotate, or AWP depending on the team and map. All this can contribute to why a distinct clustering for IGLs does not exist on the CT-side.

Moving from three to four KMeans clusters does not improve the CT interpretation. The four-cluster KMeans solution isolates a single outlier profile and leaves the remaining three clusters unchanged, so its external ARI is identical to the three-cluster solution at 0.584, while its stability falls to 0.849 ± 0.138 and its composite score falls to 0.418. The additional cluster therefore adds no role information beyond the three-cluster structure.

Overall, the CT results show that aggregated demo statistics recover a highly stable AWPer cluster and a meaningful but imperfect Rotator/Anchor split. Positional relationships and contact behavior are particularly important for distinguishing the two rifler groups, while IGL status does not emerge as an independent CT-side category.

T Side

The T-side clustering results are best summarized by the four-cluster KMeans solution. The model produced four clusters while avoiding the small, fragmented groups found in higher-k solutions. It achieved a silhouette score of 0.134, a Davies-Bouldin index of 2.140, and a composite model-selection score of 0.434. Its mean stability ARI is 0.860 ± 0.067. This stability is lower than the CT-side result, suggesting that T-side roles may be less rigidly separated in the feature space.

Against the T-specific ground-truth labels, the model reached an external ARI of 0.461 and an overall matched-label purity of 78.9%. Of the 197 eligible T-side players, 128 could be matched to the external role labels used for evaluation. The PCA visualization shows that the AWPer cluster is distinct, as expected, while the three rifler clusters overlap more substantially.

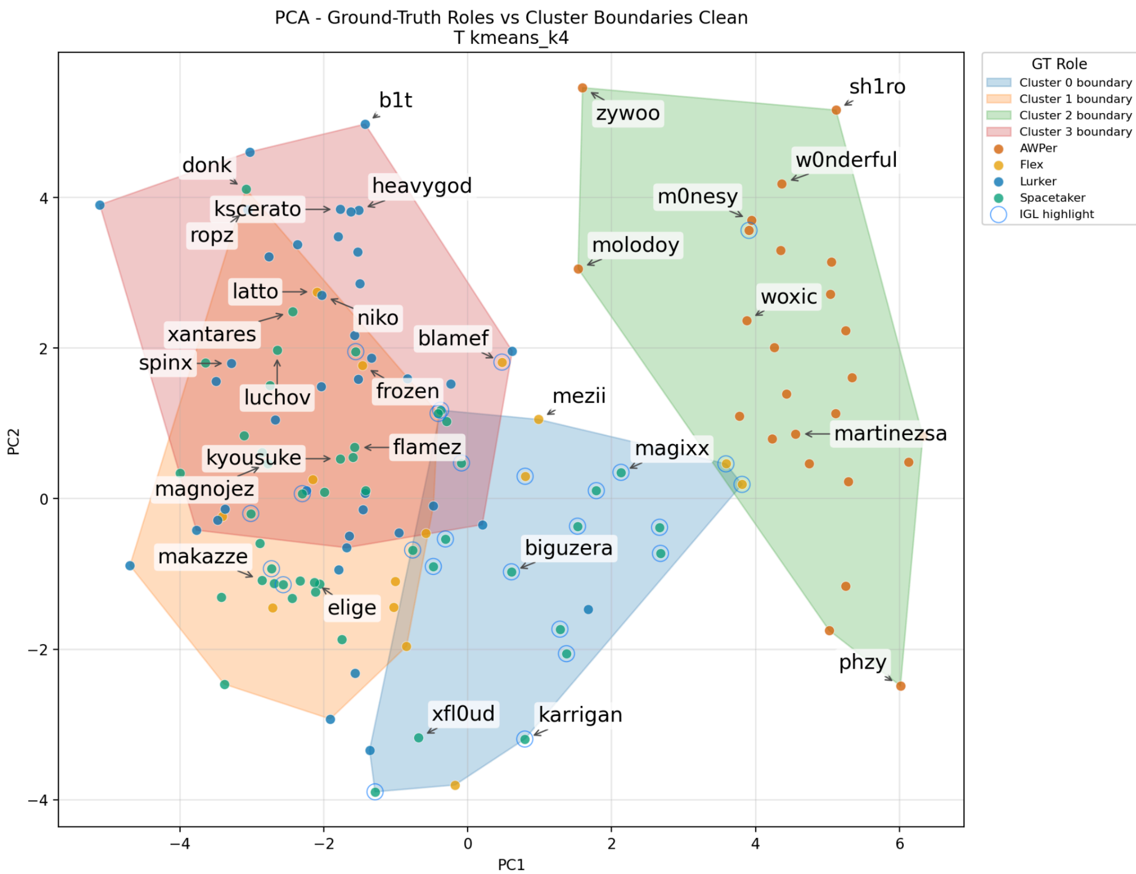


Figure 3. PCA projection of T-side kmeans_k4 clusters and external roles for matched players. Notable players are labeled.

Cluster 0: Utility-Oriented Spacetakers/IGLs, 69.6% purity. This cluster contains 43 players, with 23 matched to the external T-side labels. Sixteen of the matched players are labeled as Spacetakers. Compared with Cluster 1, these players have lower damage, kills per round, multi-kill frequency, and opening success, but use substantially more utility. The cluster therefore represents a more supportive or enabling form of space-taking rather than a purely aggressive entry profile.

This cluster also contains a strong concentration of in-game leaders (IGLs). Seventeen of its 23 matched players are labeled as IGLs in the binary IGL evaluation. This produced an IGL purity of 73.9% within the cluster. However, the overall ARI against the IGL labels is only 0.116. The model therefore identifies one IGL-enriched behavioral group, but it does not recover IGL status as a clean global division across the entire player population. This is plausible because leadership is not directly observable through demo statistics and because IGLs can perform several different roles. Furthermore, the model does not have access to team communications or strategic calls, so it can detect only the indirect effects that may be associated with leadership.

Cluster 1: Aggressive Spacetakers, 66.7% purity. This is the largest cluster, containing 70 players, of whom 45 have matched T-side labels. Thirty of those matched players are labeled as Spacetakers. The cluster is characterized by high first-contact involvement, opening kills, opening deaths, damage taken, and time spent near enemies. Players in this group also have lower survival rates and shorter average distances from opponents. These features describe aggressive riflers who frequently contest areas and create space for their teammates.

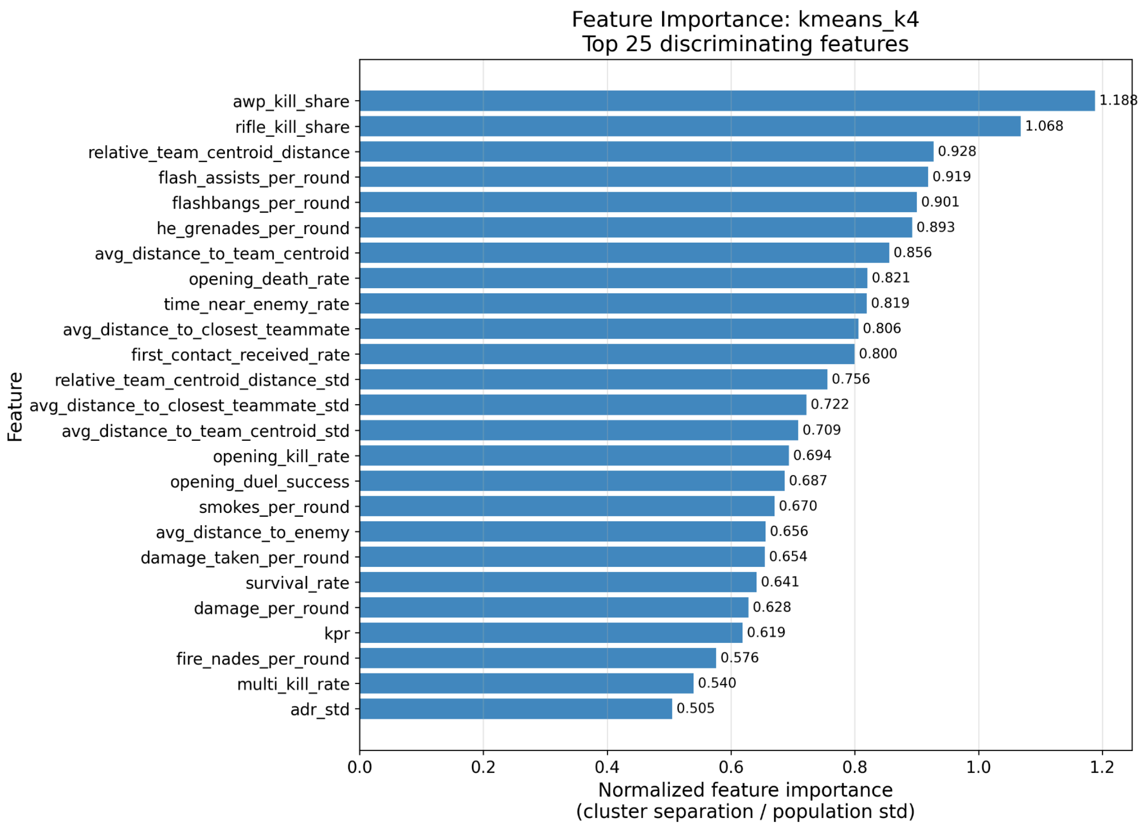


Figure 4. Top 25 features distinguishing the T-side kmeans_k4 clusters.

Cluster 2: AWPers, 100.0% purity. This cluster contains 38 players, of whom 25 have matched T-side role labels. All 25 are labeled as AWPers. The cluster has by far the highest AWP kill share and the lowest rifle kill share. Its players also tend to take less damage, die less frequently in opening engagements, and spend less time close to enemies. These characteristics reflect the longer-range and more selective engagement pattern associated with primary AWPers. As on the CT-side, weapon usage creates the clearest signal in the dataset, making the AWPer the most consistently recovered role.

Cluster 3: Lurkers, 85.7% purity. This cluster contains 46 players, with 35 matched to the T-side ground truth. Thirty of those players are labeled as Lurkers. The cluster is distinguished primarily by positioning. Players have greater average distance from their teammates and the team centroid, as well as greater variability in these positional measures across demos. This profile is consistent with lurkers, who frequently operate away from the main attacking group to control rotations, apply pressure, and exploit openings elsewhere on the map. The high purity indicates that positional isolation is a strong behavioral signal for identifying the Lurker role

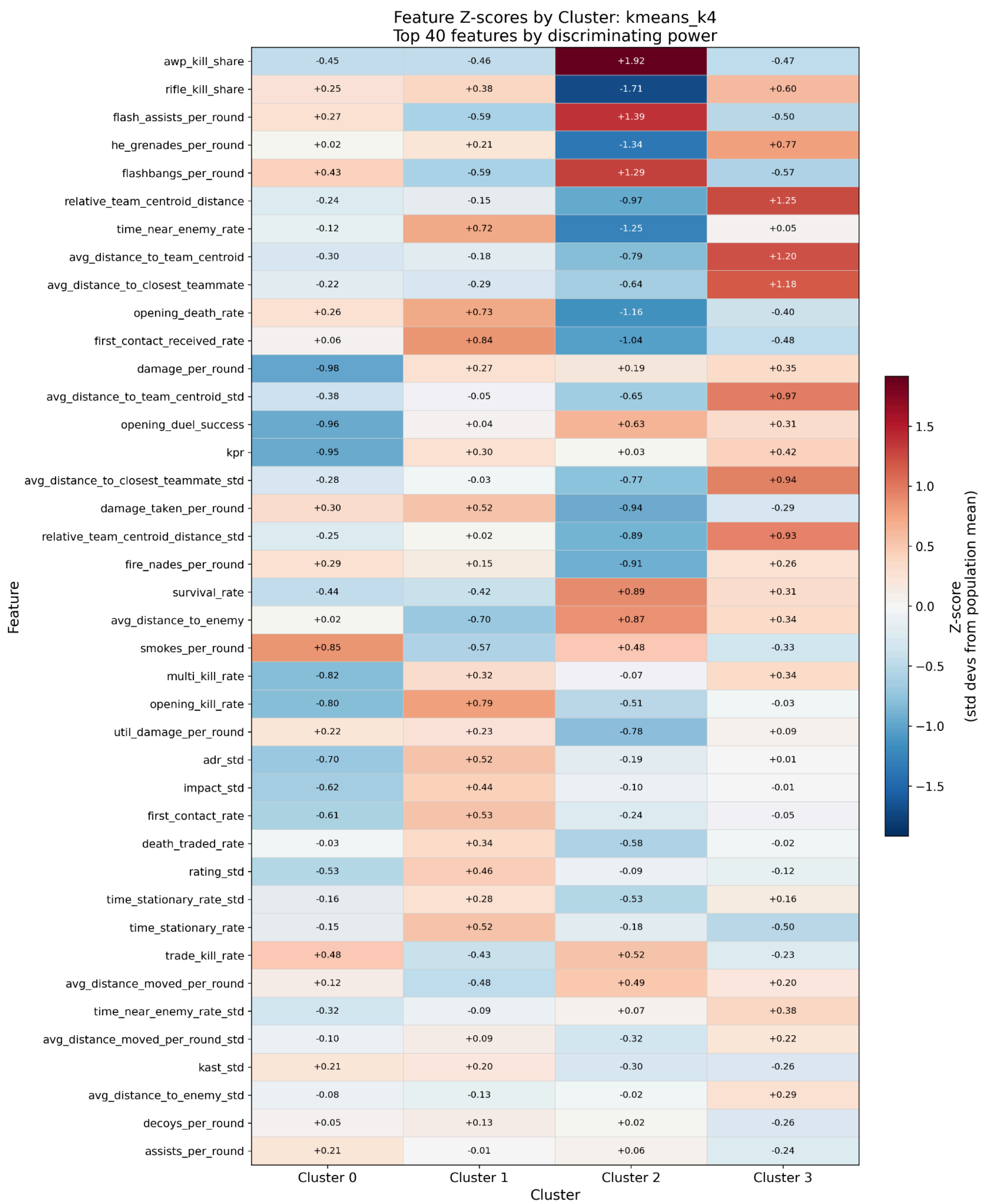


Figure 5. Standardized feature profiles for the four T-side kmeans_k4 clusters.

Overall, the T-side results indicate that meaningful role structure is present, but it is less compact and less stable than on the CT-side. The model strongly recovers AWPers and Lurkers, while dividing Spacetakers into an aggressive high-contact group and a lower-output, utility-oriented group that is enriched with IGLs. The lower stability and greater overlap between the rifler clusters are consistent with the fluidity of T-side play, where responsibilities can change according to map, strategy, spawns, and round context. For these reasons, the four-cluster KMeans solution provides the best balance of validation performance, stability, external agreement, and interpretability for the T-side.

Cross Model Observations

Across all solutions, AWPers form the cleanest clusters on both sides. AWP kill share is among the strongest distinguishing features in both models, showing that weapon specialization creates a clearer signal than most rifler behaviors.

At the selected values of k, GMM produced the same full-data partition as KMeans on the CT-side and a near-identical partition on the T-side, differing on a small number of rifler profiles (ARI 0.926), but was less stable under subsampling. Mean stability ARI falls from 0.980 ± 0.027 to 0.856 ± 0.199 on CT-side, and from 0.860 ± 0.067 to 0.695 ± 0.147 on T-side.

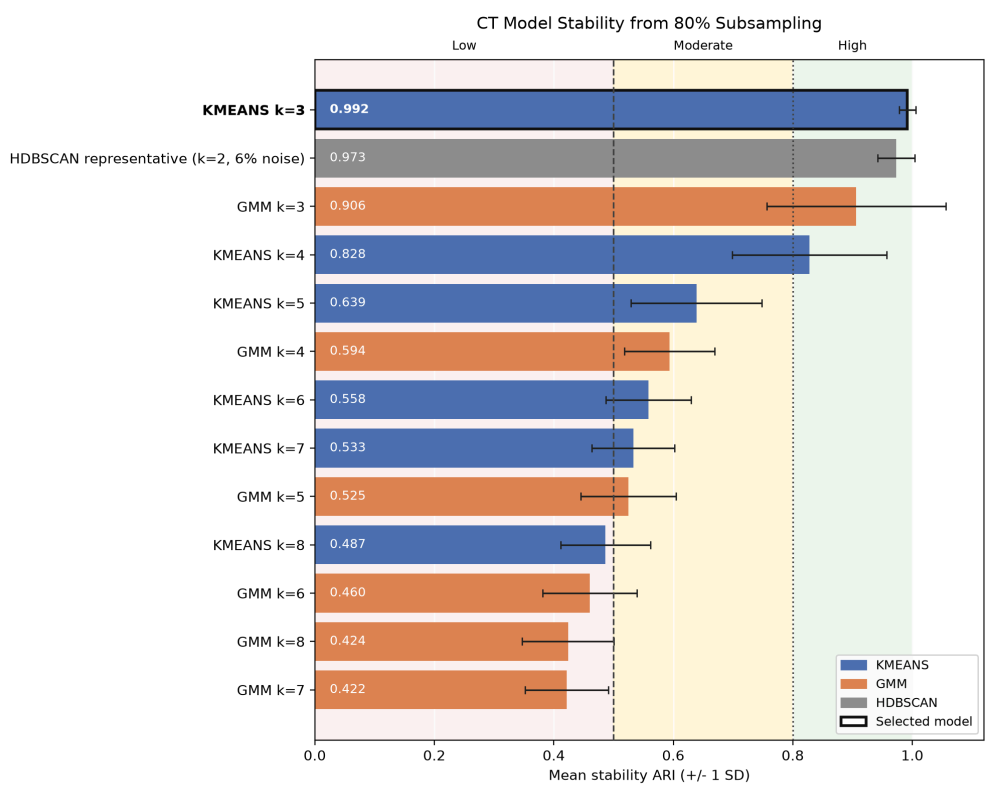


Figure 6. CT model stability under repeated 80% subsampling.

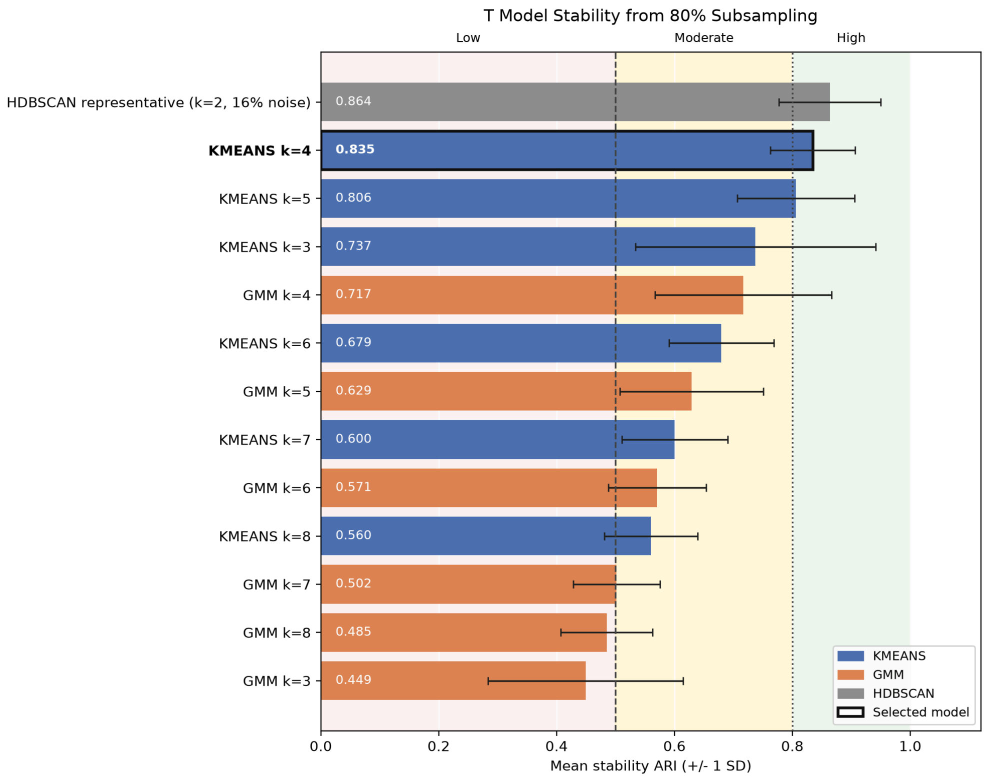


Figure 7. T model stability under repeated 80% subsampling.

HDBSCAN produced highly stable but overly broad two-cluster solutions that primarily separate the strongest weapon-profile structure from the remaining players. The leading CT configuration has a stability ARI of approximately 0.973 with 21.4% of players labeled as noise, while the leading T configuration has a stability ARI of approximately 0.913 with 22.3% labeled as noise. These solutions score well on internal metrics but do not recover the more detailed rifler roles sought by the project. This demonstrates that strong internal stability does not necessarily produce the most useful role interpretation.

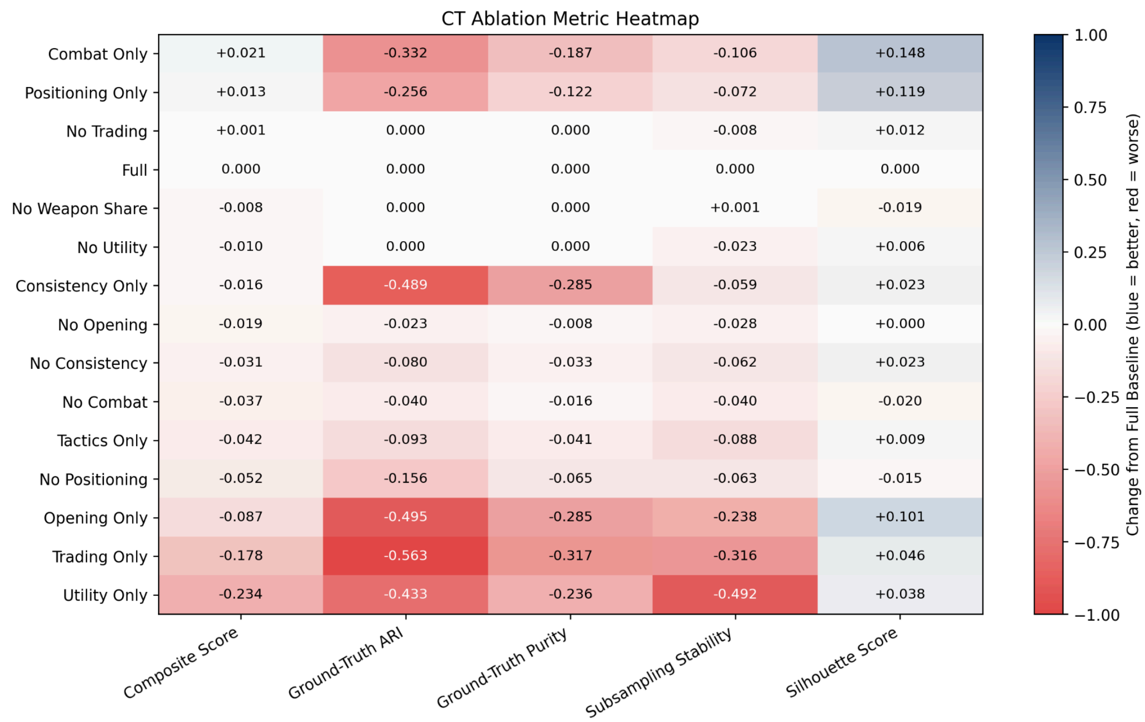
Ablation Analysis

An ablation analysis was conducted to determine which feature groups drive the CT- and T-side cluster structures. Every CT ablation was evaluated using three clusters, while every T ablation was evaluated using four clusters. Holding the number of clusters constant makes it possible to compare feature representations without allowing changes in k to create an additional source of variation.

The experiments included removal-based ablations, in which one feature category was excluded from the full model, and category-only ablations, in which clustering was performed using a single feature group. Results were evaluated using internal separation, subsampling stability, composite score, external ARI, and matched-sample purity.

CT-Side Ablations

The full CT model achieves a stability ARI of 0.992, an external ARI of 0.612, a matched-sample purity of 78.9%, and a composite score of 0.523. Removing trading and teamwork features produces nearly identical results, with an ARI of 0.612, purity of 78.9%, and a composite score of 0.524. Removing utility features also preserves the same ARI and purity, with a slightly lower composite score of 0.514. These results indicate that trading and utility variables help describe the clusters but are not required to recover the main CT structure.



	Composite Score	Ground-Truth ARI	Ground-Truth Purity	Subsampling Stability	Silhouette Score
Combat Only	+0.021	-0.332	-0.187	-0.106	+0.148
Positioning Only	+0.013	-0.256	-0.122	-0.072	+0.119
No Trading	+0.001	0.000	0.000	-0.008	+0.012
Full	0.000	0.000	0.000	0.000	0.000
No Weapon Share	-0.008	0.000	0.000	+0.001	-0.019
No Utility	-0.010	0.000	0.000	-0.023	+0.006
Consistency Only	-0.016	-0.489	-0.285	-0.059	+0.023
No Opening	-0.019	-0.023	-0.008	-0.028	+0.000
No Consistency	-0.031	-0.080	-0.033	-0.062	+0.023
No Combat	-0.037	-0.040	-0.016	-0.040	-0.020
Tactics Only	-0.042	-0.093	-0.041	-0.088	+0.009
No Positioning	-0.052	-0.156	-0.065	-0.063	-0.015
Opening Only	-0.087	-0.495	-0.285	-0.238	+0.101
Trading Only	-0.178	-0.563	-0.317	-0.316	+0.046
Utility Only	-0.234	-0.433	-0.236	-0.492	+0.038

Figure 8. CT-side ablation results with k = 3 held constant.

Removing weapon-share features also retains an external ARI of 0.612 and a purity of 78.9%. This shows that overall alignment with the external CT roles remains strong even without direct AWP and rifle kill-share variables. Other correlated features may preserve part of the same role structure, although weapon-share variables make the AWPer interpretation more direct.

The largest removal-based decline occurs when positioning and movement features are excluded. External ARI falls from 0.612 to 0.456, purity falls from 78.9% to 72.4%, and the composite score falls from 0.523 to 0.471. The clusters remain reasonably stable, but their agreement with the CT-specific role labels becomes substantially weaker. This provides the strongest evidence that the Rotator/Anchor distinction depends on spatial behavior.

This result demonstrates the main difference between Rotators and Anchors. Rotators are expected to reposition in response to mid-round information, while Anchors more often hold fixed defensive responsibilities within a site, regardless of enemy actions. Because CT-side actions are typically reactive to T-side pressure and executes, spatial behavior is more informative than raw combat or utility statistics for distinguishing the CT rifler roles.

The category-only experiments reinforce this conclusion. Positioning and movement features alone produce a stability score of 0.920 and an ARI of 0.356. Combat features alone form compact internal clusters, achieving the highest CT silhouette score of 0.314, but their ARI is only 0.280. Trading and teamwork features alone produce an ARI of 0.049, while utility features alone produce an ARI of 0.179.

Overall, the CT ablations indicate that the role structure depends primarily on positional and movement information combined with weapon and combat signals. Utility and trading features provide additional behavioral detail but have limited influence on the final external role alignment. AWPers remain identifiable through several correlated feature groups, while the Rotator/Anchor rifler split is most sensitive to the removal of spatial information.

T-Side Ablations

The full T-side model achieves a stability ARI of 0.835, an external ARI of 0.456, a matched-sample purity of 78.9%, and a composite score of 0.427. Unlike on the CT-side, no individual feature group reproduces the complete T-side clustering. Different feature categories capture different aspects of the Spacetaker, Lurker, AWPer, and IGL-enriched profiles.

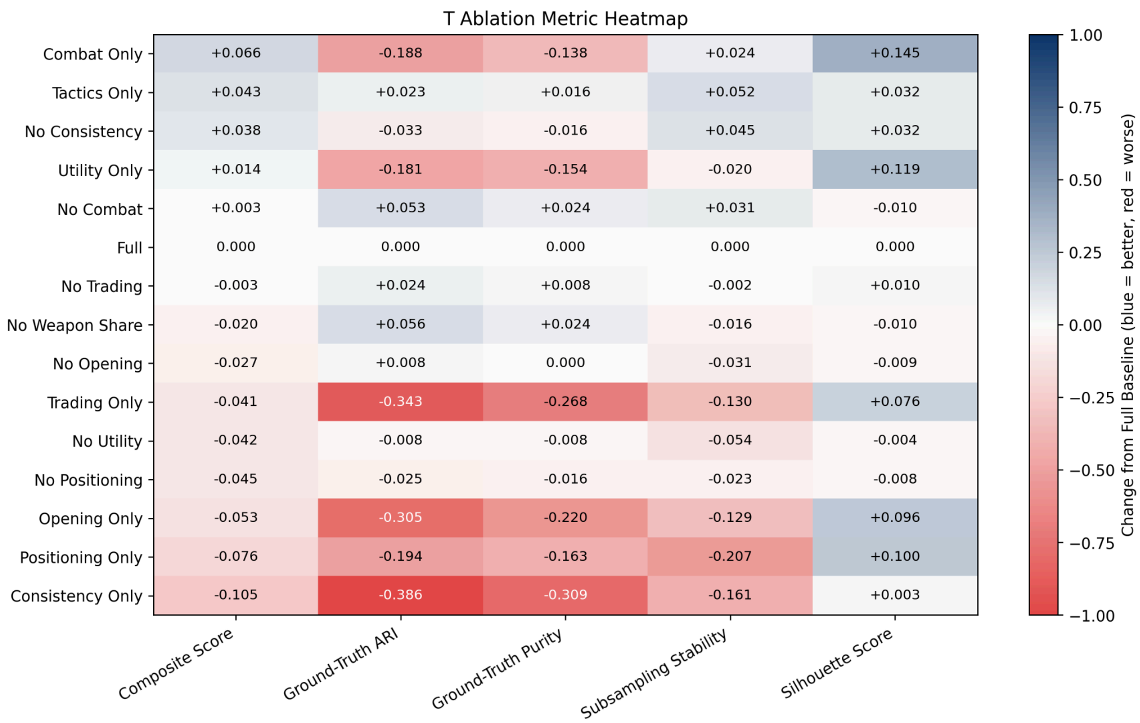


Figure 9. T-side ablation results with k = 4 held constant.

The tactics-only model combines contact, teamwork, utility, positioning, and movement features while excluding direct combat, weapon-share, and consistency variables. It achieves a stability ARI of 0.887, an external ARI of 0.479, a purity of 80.5%, and a composite score of 0.470. Its relatively strong performance shows that role information exists outside direct combat and weapon specialization. Although this model performs well, it combines several feature groups and therefore does not show that any individual feature category is sufficient.

Removing weapon-share features increases external ARI to 0.512 and purity to 81.3%, while removing combat features produces an ARI of 0.509 and the same 81.3% purity. These improvements suggest that weapon specialization may dominate the clustering, pulling players toward mechanically defined roles rather than the categories represented by the external labels. Although these ablations improve agreement with the external labels, they should not automatically be considered superior to the full model. The external annotations represent one expert-defined role framework, while the full model may preserve behavioral distinctions that are not captured by those labels.

Trading and teamwork features have limited influence on the final T-side solution. Removing them produces an external ARI of 0.480, purity of 79.7%, and stability of 0.833, all comparable to or slightly better than the full model. Trading and teamwork features alone perform poorly against the external labels, producing an ARI of 0.113. These results do not support treating trading as the primary driver of the T-side role structure.

Removing positioning and movement features reduces external ARI to 0.431, purity to 77.2%, and the composite score to 0.382. The decline is meaningful but smaller than the corresponding CT-side decline. T-side roles depend on spatial behavior, but they are not defined by positioning alone. Contact timing, opening aggression, combat behavior, weapon usage, and utility provide additional signals that preserve part of the structure when positional features are removed.

The category-only models demonstrate the difference between internal clustering quality and external role agreement. Combat-only features achieve the highest T-side composite score of 0.492 and a silhouette score of 0.283, but their external ARI is only 0.268. Utility-only features similarly produce relatively compact clusters but achieve an external ARI of only 0.274. Positioning and movement features alone produce an ARI of 0.262. These categories produce separated groups among players, but none independently reproduces the full expert-defined role structure.

Overall, the T-side ablations suggest that role structure is distributed across several feature groups. Weapon and combat variables create strong separation but can compete with broader tactical alignment. Positioning remains important, particularly for identifying Lurkers, while utility and contact-related features help characterize the lower-output, IGL-enriched Spacetaker group. The T-side results are therefore more overlapping and context-dependent than the CT-side results, showing that attacking responsibilities are expressed through combinations of behavior rather than through one dominant feature category.

Limitations

The most significant limitation is the sample size. Although the dataset contains 463 professional demos, the final clustering dataset only covers 214 players, which is small for unsupervised learning. This is amplified by the narrow player pool, which consists of top-tier Counter-Strike professionals, creating a compressed skill distribution where meaningful role differences may be subtle and overlapping.

The external role labels should also be interpreted with caution. This project uses expert curated role annotations, which provide a strong reference point, but they should still be treated as expert labels rather than absolute ground truth. Professional CS2 roles vary by map, side, opponent, economy state, roster context, and team system. Some players also occupy multiple roles simultaneously, such as an AWPer who also serves as an in-game leader. As a result, disagreement between clusters and external labels may reflect role overlap or hybrid responsibilities rather than model failure.

Another limitation is that the current feature set averages behavior across maps. This produces more stable player level profiles, but it removes important context. A player may anchor a bombsite on one map, rotate actively on another, and take different T-side responsibilities depending on the team's playbook. Map stratified clustering would likely recover more specific role patterns, especially on the CT-side, but would require a much larger sample size, as most teams in the dataset play less than 10 matches in any particular map in the current dataset.

The clustering results are also constrained by the available feature representation. The model has no access to voice communication, mid-round calls, set strategies, opponent specific preparation, or intended tactical responsibilities. This is particularly important for the IGL heavy T-side cluster. The model can only detect indirect traces of calling, such as utility usage, teammate proximity, and contact patterns. Therefore, this cluster should be interpreted as evidence that some T-side calling responsibilities may produce measurable behavioral side effects, not as proof that in-game leadership is directly observable from demo statistics alone.

Conclusion

Professional CS2 roles are partially recoverable from aggregated gameplay data. AWPers form the clearest and most stable clusters, while the model also recovers a meaningful CT-side Rotator/Anchor split and a distinct T-side Lurker group. Positional features are particularly important on the CT side, while T-side roles depend on combinations of combat, utility, contact, and positional behavior.

Flexible rifler and leadership responsibilities remain difficult to separate because they vary across maps, teams, and tactical contexts. The IGL-enriched T-side cluster therefore represents a measurable behavioral overlap rather than direct identification of in-game leadership. Future map-stratified analysis could preserve positional structure currently lost through aggregation.

References

[1] Xenopoulos, P., and Awpy contributors. (n.d.). Awpy: Counter-Strike 2 demo parsing, analytics, and visualization in Python [Computer software]. Retrieved August 6, 2026, from https://awpy.readthedocs.io/en/stable/

[2] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830. https://www.jmlr.org/papers/v12/pedregosa11a.html

[3] McInnes, L., Healy, J., and Astels, S. (2017). hdbscan: Hierarchical density based clustering. Journal of Open Source Software, 2(11), 205. https://doi.org/10.21105/joss.00205

[4] Richards, H. (n.d.). Positions Database NER0cs [Tableau dashboard]. Tableau Public. Retrieved August 6, 2026, from https://public.tableau.com/app/profile/harry.richards4213/viz/PositionsDatabaseNER0cs/PositionsDatabaseNER0cs

[5] HLTV.org. (n.d.). HLTV.org: The home of competitive Counter-Strike. Retrieved August 6, 2026, from https://www.hltv.org/